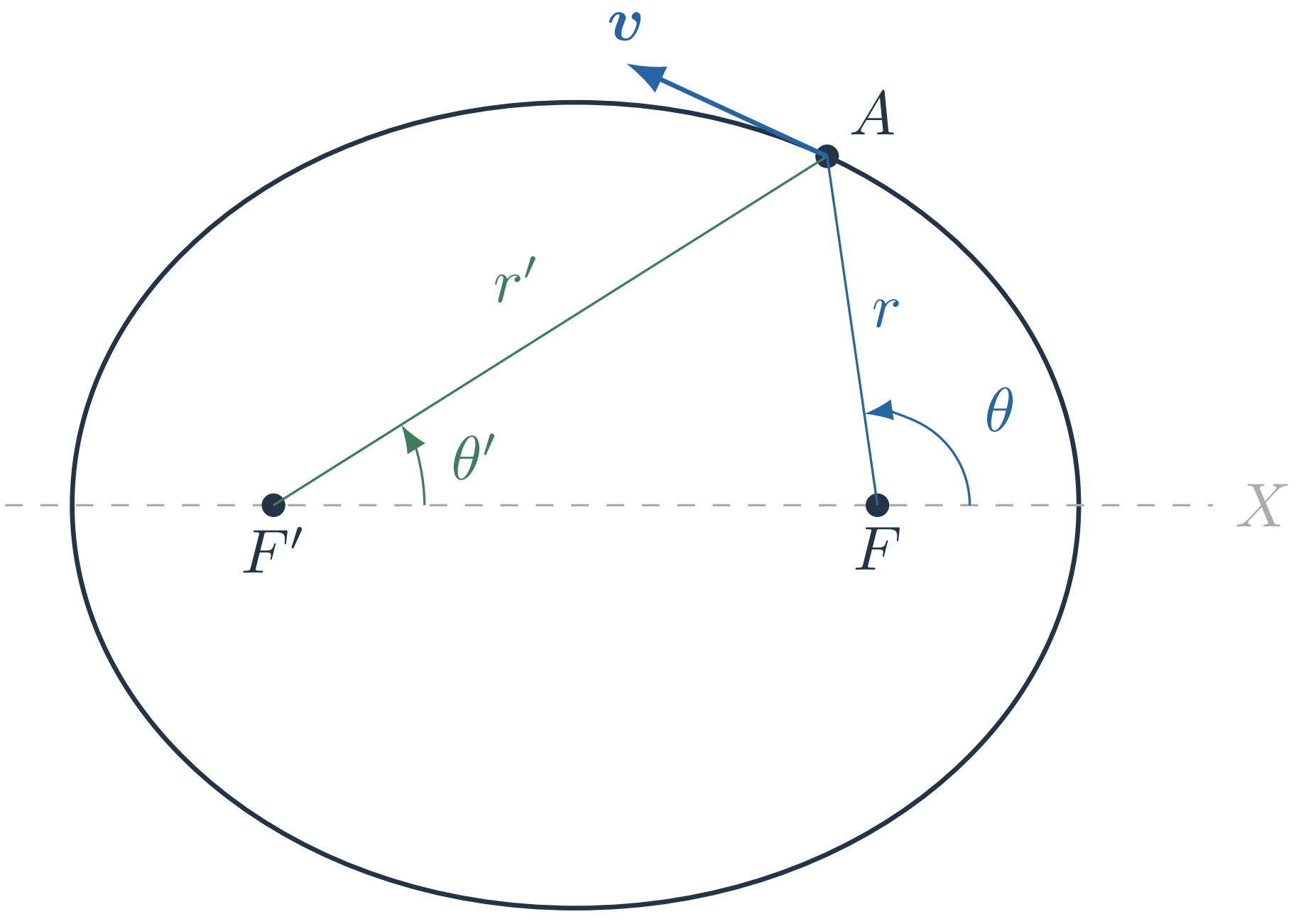
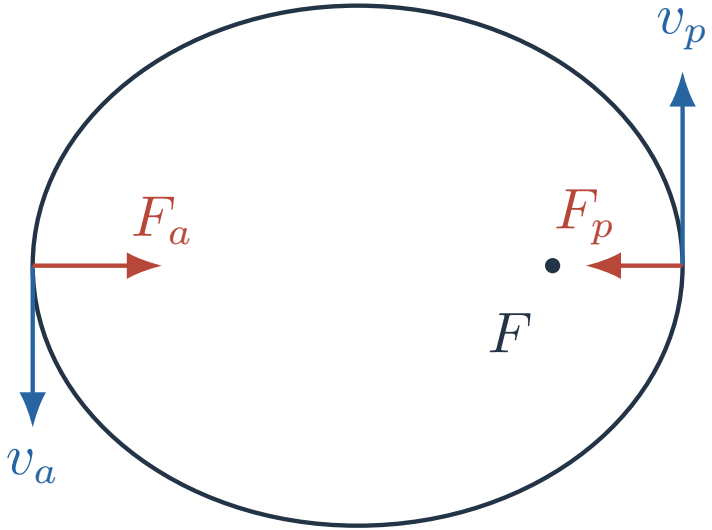
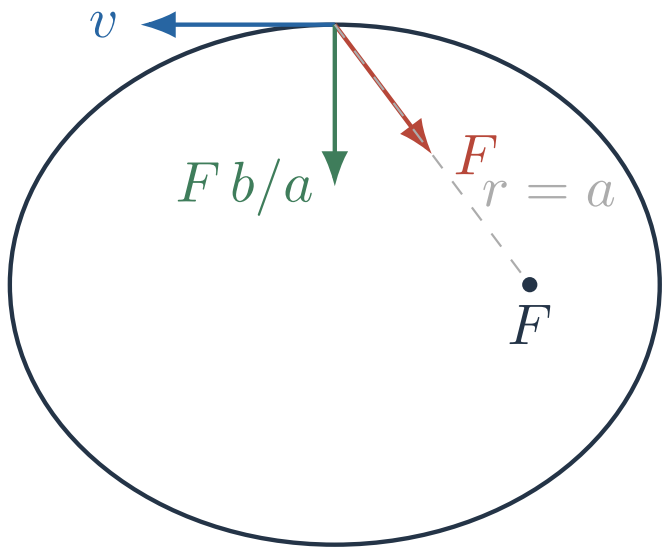


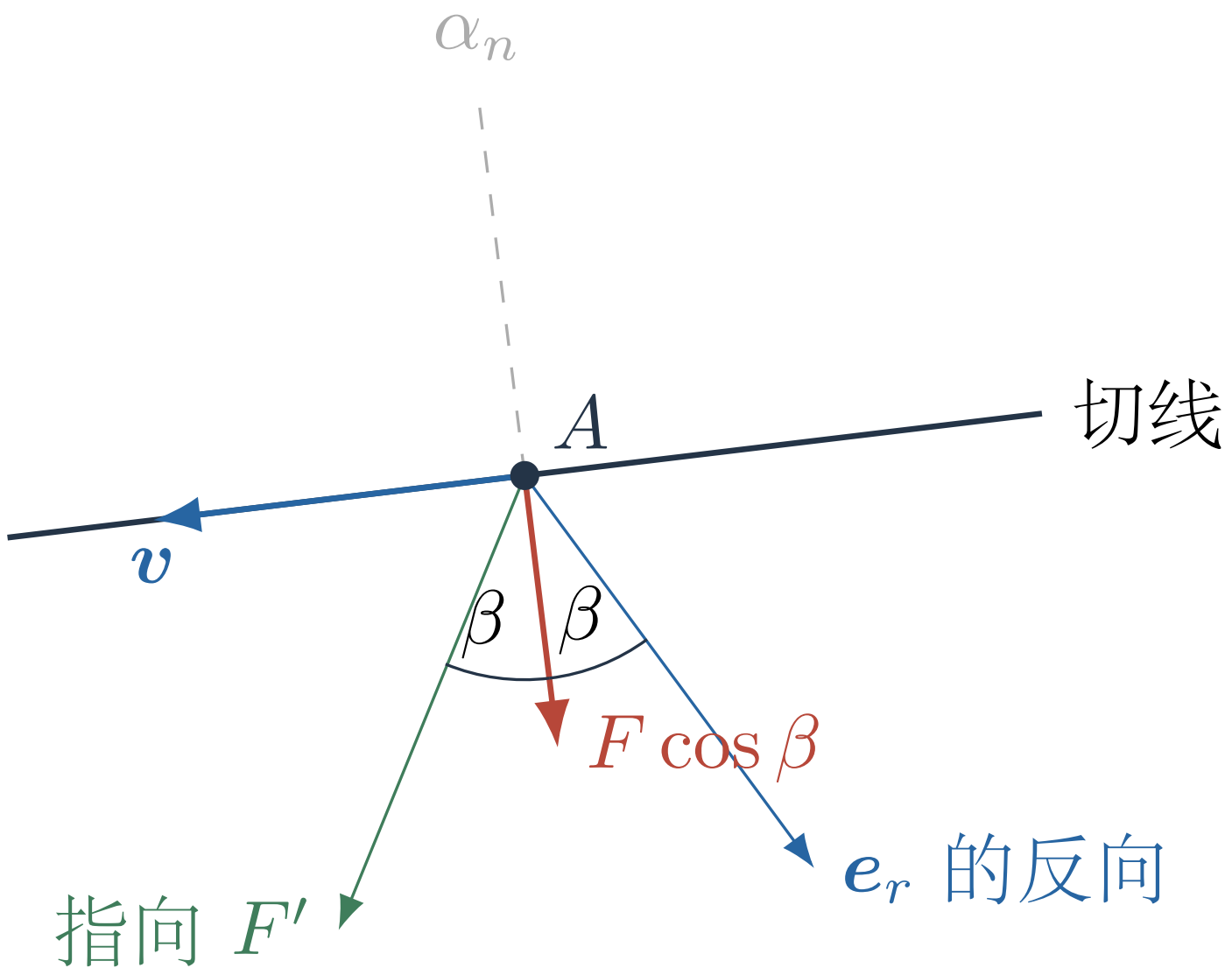
焦点间距 $2ae$；逆时针运动，$r + r' = 2a$



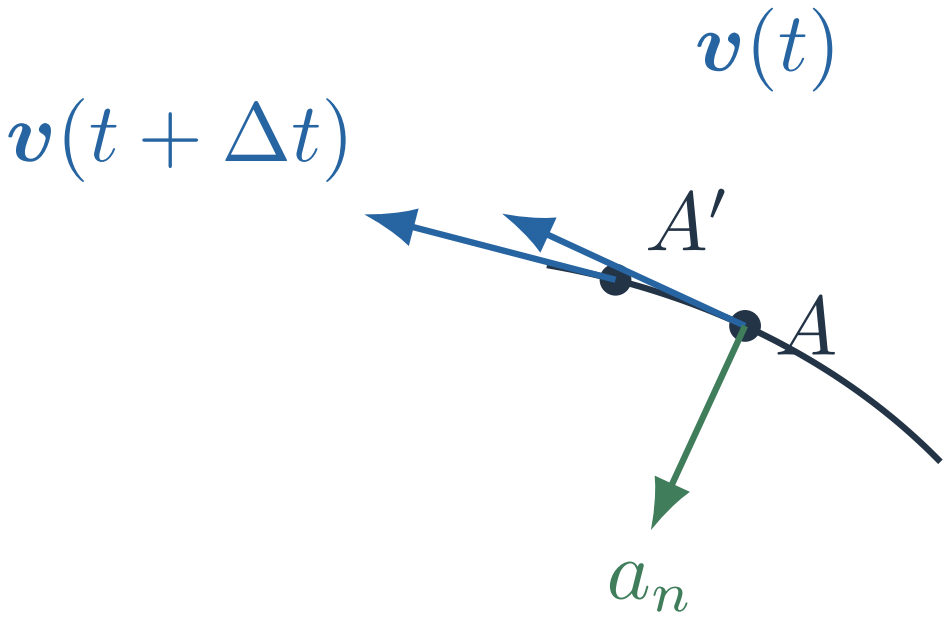
长轴两顶点：$\rho = l$，中心力即法向力

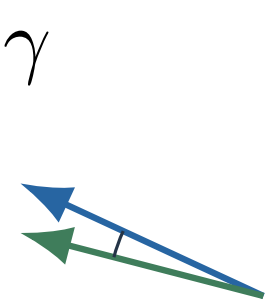


短轴顶点：$\rho = a^2/b$，需投影



法线平分两焦点方向；中心力与内法线夹角为 β





$$\Delta\gamma = \frac{v\Delta t}{l}\cos^3\beta$$

相邻切线转角的示意；计算取 $\Delta t \to 0$